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Magnetron sputtered thin film transparent conductive coating for solar cell application

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Transparent Conductive (TC) Coating is mainly based on fluorine doped tin oxide (SnO_2:F), indium-tin oxide (ITO) and doped zinc oxide (ZnO:Al, ZnO:Ga, ZnO:B etc.) *thin films, ultrathin metal film embedded in two dielectric layers* on glass and plastic substrates. Graphene based material is important alternative member in transparent conductor (TC) family for substitution of existing TCO materials. *T*extured *TC coating* with sheet resistance < 5 Ω/□, > 90% optical transmittance and higher figure of merits plays key role for opto-electronic device applications. Textured TCO used as *transparent window as well as* front *electrodes* and back reflector *applications* for thin film *solar cells*. Surface texture (in terms of Haze factor) of TCO films controls the light scattering and plays crucial role for improvement of effective light absorption in the active layer of thin film multijunction solar cell. *Magnetron sputtering technique fulfills all necessary criteria to achieve unique* materials properties *for commercial TCO thin film development.*

Keywords: Magnetron sputtering, doped-ZnO thin film, electrical and optical properties, haze factor and scattering mechanism, work function, graphene thin film.

1. Introduction

1.1. Fundamentals of sputtering

Sputtering is widely used industrial technique for vapour phase thin-film growth in the field of surface coating technology mainly for hard coating, decorative coating, optical

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coating, transparent conducting oxide coating, and coatings of refractory compounds, metals and semiconductor materials. Rf-magnetron sputtering offers low temperature deposition mechanism that provides high quality uniform homogeneous ion damage free dense surface coating with better thickness control, high deposition rate and high sticking co-efficient over large area on the substrate surface.

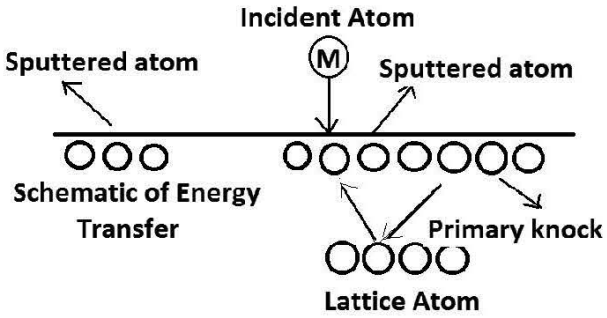


Fig. 1. Ion Bombardment during sputtering.

Sputtering means the ejection of solid atoms due to momentum transfer by the bombardment of gas ions (+ve ions) from solid target in a high vacuum environment and finally deposited on a substrate surface to develop thin film coating. It is one of the important high vacuum plasma deposition processes for thin film fabrication by producing capacitively or inductively coupled plasma. During sputtering at low pressure environment and chamber heating arrangement, contamination free coating can be developed with the removal of all traces of adsorbed contaminant gases from chamber wall and target material through desorption. The necessary conditions for self sustaining plasma, chamber geometry plays significant role, chamber pressure will be around 3 to few hundred mTorr and sufficient secondary emission by the electrons maintain plasma densities around 10^{10} - 10^{12} cm^{-3}. Higher chamber pressure offers more collision and low mean free path between positive ions and sputtered atoms. Maximum energy transfer in such a collision occurs when the masses are equal.

During ions bombardment on target surface several things like reflection, sticking (adsorption) sputtering, ion implanta-

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tion, chemical reactions, and electron - photon emission occur depending upon ion beam energy at the time of ion-surface interactions. Sputtering process is associated with energy transfer (Elastic and inelastic energy transfer) and for the collision between the particles having equal masses shows maximum energy transfer. Elastic energy transfer depends on mass as well as angular distribution but during inelastic energy transfer secondary electron emission occurs due to ions - electrons interaction on the target material. These "secondary electrons" are then accelerated towards the anode and participate in plasma generation by ionizing the sputtering gas that will play leading role for sustaining Plasma.

The ion beam energy is the critical parameter

< 5 eV : Adsorption or reflection

5 – 10 eV : Surface damage and migration

10 – 3 keV : Sputtering

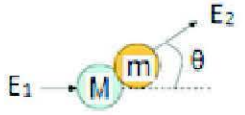
> 10 keV : Ion implantation

Sputter Yield (S) is the key factor for sputtering process defined as the ratio between the numbers of ejected atoms to the total numbers of incident atoms and is given by

$$S = \frac{\text{Number of Ejected Atoms}}{\text{Number of incident Atoms}} = \alpha \frac{Mm}{(M+m)^2} \frac{E_m}{U_M}$$

Where, M : mass of target atom
m : mass of bombarding ion
E_m : kinetic energy of bombarding ion
U_M : Bonding energy of target metal
α : depends on striking/incident angle

Here, energy transfer during Ion-Target surface interaction follows the given relation (considering oblique collision)



$$\frac{E_1}{E_2} \alpha \frac{Mm}{(M+m)^2} \cos^2\theta$$

S can range from 0.1 to 10. S depends on type of target atom, binding energy of target atoms, relative mass of ions, incident ion energy, and angle of incidence of ions etc. Angular distribution of sputtering depends on the chamber pressure.

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- Lower pressures result in a more directed flow produces less uniform films.
- Higher pressures result in more isotropic flow and better coverage.
- Uniform films also require larger targets.

Ultra-Low pressure of the order 10^{-6} pa is artificially created by cryogenic pumps or turbo-molecular pumping system. Chamber heating arrangement helps for desorption of adsorbed hydrogen, or oxygen molecules from chamber wall. Sputtering process can be classified into reactive as well as non-reactive sputtering. In general Ar gas is introduced for plasma production. Refractory compound materials are deposited in presence of some reactant gases like hydrocarbon, nitrogen, or oxygen, hydrogen etc. During reactive sputtering, sometimes sputtering target becomes contaminated due to the reaction of reactive gas with exposed part of target materials. Pre-sputtering and re-evacuation to the base vacuum (of the order 10^{-6} pa) is one an important solution contamination free thin film growth.

Several arrangement and modifications can be implemented in modern sputtering system.

- ***DC sputtering*** is the simplest and low energy efficient technique. It requires an electrically conductive target for thin film coating; insulator coating deposition is restricted due to plasma sheath formation near the target surface. Electron bombardment may cause significant damage of the substrate as well as coating surface.
- ***Radio-Frequency (RF)*** sputtering is very much useful for insulators coating deposition in the frequency range 13.56 MHz, with slower deposition rates.
- ***Magnetron sputtering*** is basically made up with magnetically enhanced cathode (magnetron) (*i.e. static magnetic field is configured in the cathode location*). Secondary electrons, generated by ion bombardment are constrained to flow perpendicular to the E x B. That means magnetically confined plasma is produced with electrons trajectory into spiral paths to increase collision frequency with gaseous neutrals. In the correct configuration, the secondary electrons follow a current loop called the "drift ring" parallel to the cathode surface and faster ionization takes place near the target

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surface. The extra Ar ions created by collisions offers faster growth rates and also plasma can sustain at lower chamber pressure. Charge neutrality of sputtered atoms makes them undeviated from magnetic trap. Charge deflection mechanism under electro-magnetic field additionally provides ion damage free film. Magnetron sputtering also has some unique features in chamber design for multi-target arrangement in closed filed magnetron sputtering and unbalanced magnetron sputtering system. Rf-Magnetron sputtering is very much effective for electron/ion-damage free TCO film development for high deposition rate, fine control of thickness over large area & uniformity, surface texture etc.

- ***Unbalanced magnetron sputtering*** is a relatively modified version of glow discharge as magnetron sputtering processes for thin film deposition. In *unbalanced magnetron sputtering* thin film coating is deposited on proper substrate with a cylindrical target. A significant reduction of ignition and glow discharge voltages were observed by adding magnets to carry out the *unbalanced magnetron sputtering*. Sputtered films have many important technological applications mainly for surface passivation by insulating and diffusion barrier (SiN_X, SiO_2 etc.) coating in semiconductor devices, hard coating (TiN, SiTiC thin films) for cutting tools and erosion applications. Many other coating materials like TaN resistive films for hybrid circuits, W, Cu, Al films for metallization of semiconductor devices are deposited by unbalanced magnetron sputtering.

1.2. Transparent conducting coating

Doped tin oxide (SnO_2), indium tin oxide (ITO), and zinc oxide (ZnO) are the important members of TCO family, dominated TCO market and mass-production is going on during last three to four decades to fulfil the market demand. During last 5 years. Considering many basic research issues in laboratory R & D large scale manufacturer needs careful handling during modification TCO materials characteristics. Doping is a critical factor to control electrical characteristics as well as optical transmittance. Resistivity of the material can be controlled by either with increase of carrier concentration or mobility or optimizing both. With increasing of carrier density, resistivity decreases but leads to an increase in the visible

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absorption, but increase of carrier mobility don't have any adverse effect on optical properties. The coexistence of electrical conductivity and optical transparency in these materials depends on the nature, number and atomic arrangements of metal cations in crystalline oxide structures, the presence of intrinsic or intentionally introduced defects, and on the resident morphology and on the proper surface texture etc. High work function of TCO films is another sensitive parameter suitable for new generation organic/ inorganic hybrid and metal organic perovskite based solar cell applications.

Transparent conductive oxide (TCO) films are becoming very much demanding materials for its rapidly growing such as flat panel displays (FPD), liquid crystal displays (LCD), Thin Film Transistor (TFT) and solar cells. Continuous development of the high performance optoelectronic devices stimulates the research on TCO films. For TCO films in solar cell applications, high conductivity and high transparency in the visible and near-infrared (NIR) wavelengths are required in order to increase conversion efficiency and stability of solar cells[1-7].

The important TCO semiconductors are impurity-doped ZnO, In_2O_3, SnO_2 and CdO, the ternary compounds Zn_2SnO_4, $ZnSnO_3$, $Zn_2In_2O_5$, $Zn_3In_2O_6$, In_2SnO_4, $CdSnO_3$, and multi-component oxides consisting of combinations of ZnO, In_2O_3 and SnO_2. Sn doped In_2O_3 (ITO) and F doped SnO_2 TCO thin solid films are the most preferable materials for most present applications. The most preferable TCO materials are ITO, AZO, CdO, GZO mainly used as a transparent front electrode and back reflector in thin film Solar cell for enhancing its power conversion efficiency as well as its stability. Conventional transparent conducting oxides such as ITO, SnO_2:F are not stable under hydrogen plasma atmosphere, hence degrades the optical transparency under high hydrogen plasma atmosphere. The problem regarding ITO films: Indium is a rare metal, typically found in limited geographical areas like China and Indium is very costly material. With respect to the present scenario indium will soon be on the strategic resource lists of every country in the world. This situation drives the search for alternative TCO materials to replace ITO. Therefore new alternative technology based on ZnO thin film need for replacement.

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For establishment of ZnO based TCO technology the following problems will be solved stated below:

1. Simultaneous high optical transmittance and electrical conductivity i.e higher figure of merit ($\phi_{TC} = T^{10}/R_{sh}$),
2. Work function equal or greater than 5 eV,
3. Uniformity.
4. For ultra-thin ZnO thin film (film thickness < 300 nm)
   (a) Film resistivity (ρ) $\sim 10^{-5}$ ohm-cm,
   (b) Sheet resistance (R_{sh}) < 5 ohm/square.
5. Control of Surface Texture and proper Haze ratio is key factor,

Therefore new alternative technology based on ZnO thin film need for replacement because ZnO having some advantageous are

1. Zinc oxide (II–VI group) is an n-type semiconductor with the structure of hexagonal wurtzite and a direct allowed gap with a band gap of 3.3 eV at room temperature[8]
2. Relatively Large exciton binding energy (60 meV)
3. Large Piezoelectric constant.
4. Strong luminescence
5. High thermal conductivity
6. Amenability to wet chemical etching
7. Radiation hardness
8. Strong sensitivity of surface conductivity to the presence of adsorbed species
9. Chemical stability in hydrogen plasma, thermal stability when doped with group III elements abundance in nature, and no toxicity[1–8].

With these properties, there are very large fields of application in industrial and scientific research, particularly for both microelectronic and optoelectronic devices. Zinc oxide is applied in transparent conductors, solar cell windows, heat mirrors, gas sensors, and surface acoustic wave devices[9–14].

There are several techniques for producing ZnO semiconductor thin films. These techniques are molecular beam epitaxy (MBE), chemical vapour deposition (CVD), vacuum

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evaporation (VE), sputtering, pulsed laser deposition (PLD), electrochemical deposition, and spray pyrolysis methods (SP)[15–22]. However transparent conducting ZnO thin films are preferably deposited by rf-magnetron sputtering due to some unique advantages compared to the other process (such as for electron/ion-damage free TCO film development for high deposition, fine control of thinness over large area and uniformity, surface texture).

The growth technique and chamber design play a significant role in controlling the different materials properties of ZnO films. The electrical and optical properties of the films are strongly dependent on the structure, morphology, nature of dopant and also due to the variation of the deposition parameters and chamber geometry. Therefore, detailed investigations on different TCO films deposition techniques are very much important.

At present, ZnO:Al, ZnO:Ga (AZO and GZO), IZO:Ga semiconducting thin films prepared by Pulsed Laser Deposition (PLD) are most promising alternatives to ITO thin-film with novel electrical (Resistivity $\sim 10^{-5}$ ohm-cm) and optical properties[1,2]. Das *et al.* have reported detail material properties of Al-doped ZnO films prepared by rf-magnetron sputtering deposition with resistivities of the order of 2.5×10^{-4} ohm-cm[3].

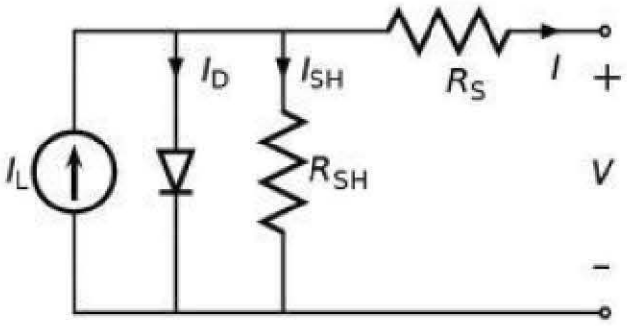


Fig. 2. Equivalent circuit of a solar cell, R_s and R_{sh} is lumped as an equivalent series and shunt resistance.

1.3. Basics of solar cell

Solar cell is basically a p-n junction diode that converts light energy to electricity. Under illumination, sunlight is absorbed in the active layer of solar cell, and photo-generated 'electron

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and hole' pairs are produced. Electrons are collected in n-side and holes are collected in p-side, as a result potential difference is build up across two electrodes (known as *Barrier Potential*). These electrons and holes are separated across the junction and provide the electrical potential to drive them through an external circuit as current. This is the basic mechanism to draw the output power by Solar cell. In a real solar cell some power is lost for driving current through the inactive portion in the solar cell.

Also sheet resistance of the layer and the contact resistance cause some dissipation of power. These losses are lumped as an equivalent series resistance R_S. The current-voltage (I-V) behaviour of an ideal p-n junction solar cell is given by

$$I = I_0\left\{\exp\left[\frac{eV}{nkT}\right]-1\right\}-I_L$$

Where, I_0 determines the reverse saturation current the characteristics. Cancelling of *ID (dark current)* by means of I_L can be considered by setting $I = 0$. I-V Characteristic curve and Solar photovoltaic parameters are depicted in the Fig. 3 (shown below).

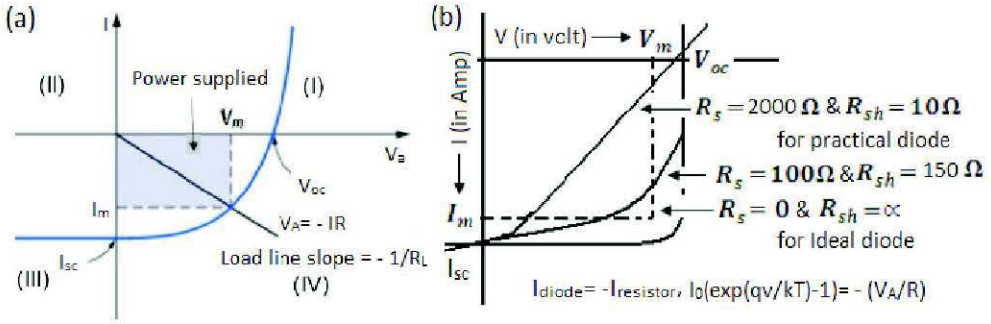


Fig. 3. (a) I-V Characteristics of Solar Cell, (b) Series and Shunt Resistance of Ideal and Practical Solar Cell.

Photovoltaic Parameters of Solar Cell

Open Circuit Voltage (V_{oc}): $V_{oc} = (nkT/q)\ln(I_L/I_0+1)$

Maximum Power output (P_{max}): $P_{max} = I_m V_m$, Fill Factor (F.F): $F.F = (I_m V_m)/(I_{sc} V_{oc})$

Power Conversion Efficiency (η) $\eta = P_{out}/P_{in} = (I_m V_m)/P_{in} = (FF.I_L.V_{oc})/P_{in}$

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In I-V characteristics for different R_s and R_{sh} are shown in above Figure. For single junction Si solar cell of 2 cm^2 areas the two important PV parameters are mainly given by V_{oc} ~0.5–0.6 V and I_{sc} ~30–60 mA. In solar panel several cells are connected in series and parallel and finally voltage and finally current of the PV module is increased. Ideal conversion efficiency of a solar cell is defined as the ratio between the outputs to input power, given by

$$\eta \propto (n_{ph})_{E_g} . E_m$$

$$\eta = \frac{P_m}{P_{in}} \times 100\%,\ P_{in} = A\int_0^{\infty} n_{ph} d\,(h\nu)$$

Highest achieved value of power conversion efficiency (η) is 31% for a semiconductor of E_g = 1.35 eV and of spectrum AM 1.5. This is the highest value of efficiency achievable with a single p-n junction diode solar cell. Energy band gap (Eg) values for some common element and compound semiconductor used in solar cell are given below in Table 1.

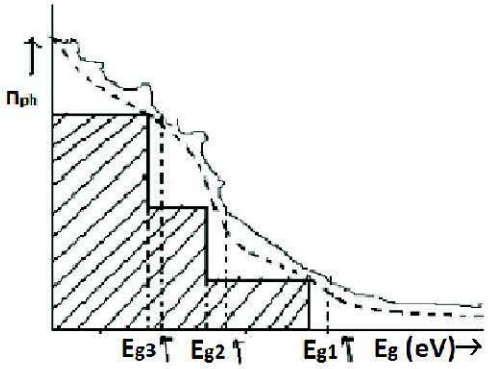
Table 1. Important solar cell materials and latest performance of different types of solar cell

Solar cell materials	Solar cell type	Lab-scale % efficiency	Module efficiency (%)
c-Si, 1.12eV	Si-single crystal	24.8	17
a-Si, 1.78eV	Si-HIT single	21.5	16.5
Ge, 0.67eV	Si-Polycrystal	20	16
GaAs, 1.42eV	Amorphous Si	13	20
CdTe, 0.56eV	Thin film CuInGaSe	20.3	29
CIGS, 1.7eV	Thin film CdTe	19	29
InP, 1.35eV	CuZnTeSe	9.6	30
AISb, 1.5eV	GaInP/GaAs	30.3	30.3
CdS, 2.42eV	Gratezel	11	5
Organic polymer	Multijunction	45.7	Not known
Organic dye	Tandem polymer	10.6	Not Known

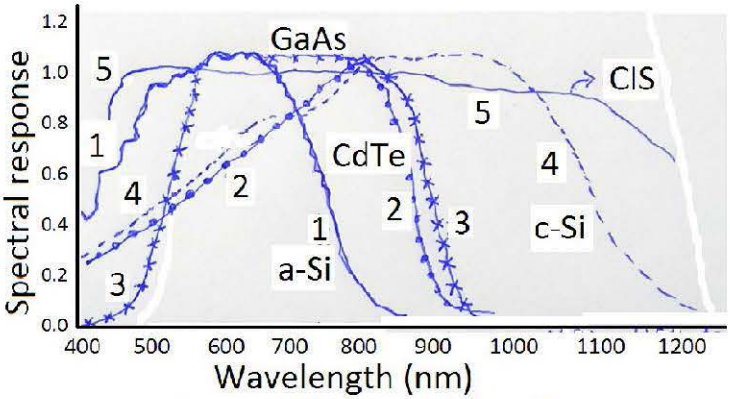
ower conversion efficiency of solar cell may be increased by using multiple p-n junctions with different values of band gap (E_g) combinations. With three p-n junctions of Eg_1, Eg_2 and

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Eg_3 the ideal conversion efficiency is obtained from the shaded area shown in the figure below. For $Eg_1 > Eg_2 > Eg_3$, the sunlight is absorbed successfully in Eg_1, Eg_2, Eg_3. Greater portion of solar spectrum will thus be used for production of electrical power. J. Yang *et al.* in NREL lab prepared an amorphous-Si-based alloy triple-junction solar cell with 14.6% initial and 13.0% stabilized power conversion efficiency. To achieve the above result low band-gap amorphous silicon–germanium alloy based p-n tunnel junction was used in between the component cells, and the top conducting oxide.



Solar Spectrum Utilization in Multijunction Solar Cell



Quantum Efficiency of Solar Cell

Fig. 4

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Solar cell materials and power conversion efficiency

Over the past half century, mainly Si-/GaAs-based crystalline solar cells were used widely but various type of thin film solar cells especially (CdTe, CIGS, DSSC, organic solar cell etc.) have been developed, which are most efficient and cost effective.

According to the type of photo voltaic material used, solar cells are classified into following categories:

1) Crystalline Si- Solar Cell
2) Amorphous silicon (a-Si) and other thin-film silicon (TF-Si),
3) Cadmium telluride (CdTe) (using binary materials),
4) Copper indium gallium diselenide (CIS or CIGS), (using ternary/alloy semiconductor),
5) Dye/Quantum dot sensitized solar cell (DSC),
6) Organic solar cells and
7) Organic-Inorganic hybrid solar cell.
8) Perovskite Solar cell.

2. Growth kinetics of magnetron sputtered doped-ZnO thin films

2.1. Effect of process parameters

Aluminum doped zinc oxide films (ZnO:Al) thin films were prepared under different gas ambient by RF-Magnetron sputtering on glass substrate at room temperature as well as higher substrate temperature using ZnO:Al target with 98:2 wt% ZnO and Al_2O_3 ratio. In general, Ar is used as sputtering gas during sputtering. Das *et al.*[23] used O_2 and H_2 gas with Ar during sputtering and detail study on thin film growth kinetics was reported. Introduction of H_2 and O_2 with Ar make plasma diluter and slow down growth rate for optimum chamber pressure and sputtering power. At RF-Power 100 watt, chamber pressure at 3m Torr, substrate temperature T_s = 300°C deposition rate increases first with increase of Ar flow rate, but excessive increase of gas flow rates increases chamber pressure. At high chamber pressure the scattering probability of precursor ions increases and makes more collisions and as a result ultimately the growth rates ZnO film decreases. The deposition rate of Ar deposited ZnO:Al films increases from 20 nm/min to 70 nm/min. On the other hand, deposition rate of Ar

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deposited ZnO:Al films decreases from 80 nm/min to 30 nm/min. Similar trends were observed in both RF-power and pressure variation on growth rates for $Ar+O_2$ and $Ar+H_2$ deposited ZnO:Al films. During reactive sputtering under $Ar+O_2$ and $Ar+H_2$ plasma, Ar plasma effectively becomes diluted. Reaction kinetics and momentum transfer in different plasma atmosphere becomes different for different precursors.

Water vapour plays important role to control stoichiometry, internal texture as well as surface texture of ZnO:Al thin film. Nakada *et al.*[24] reported the influence of water vapour on the properties of DC-Magnetron sputtered 2-μm-thick ZnO:Al films. ZnO:Al films were developed using H_2O-Ar(1:1) as sputtering gas at 300–400°C substrate temperature by DC magnetron sputtering and reported 9 ohm/square sheet resistance along with 80% optical transmittance at 550 nm after vacuum annealing.

Sputtered sample is denser, compact and highly crystalline due to more kinetic energy transfer from highly energised sputtered species to the substrate surface. During early stages of film formation Zn atoms bind to the O-terminated surface and free O ad-atoms or O_2 dimers binds with Zn^{23}. In this way, ZnO crystallites are formed on the substrate surface, and clusters are formed. The ad-atom cluster can start to diffuse across the surface. When the clusters reach to the appropriate site, Zn atom can splits from the cluster.

2.2 Effect of substrate materials and substrate temperature

Deposition mechanism is highly dependent on substrate materials and substrate temperature. Growth kinetics depends on nucleation as well as lattice matching with substrate materials. Substrate temperature plays crucial role for growth rates of TCO thin films.

During thin film materials development adsorption, chemisorptions, surface diffusion, nucleation and growth are followed for uniform dense thin film development (steps are shown in Fig. 5). Choice of substrate materials for specific application of TCO coating, especially transparent polymer is very good selection as substrate material for flexible electronics application. Adhesion of coating materials with substrate surface is really an important issue and surface functionalization of organic polymer with TCO molecules

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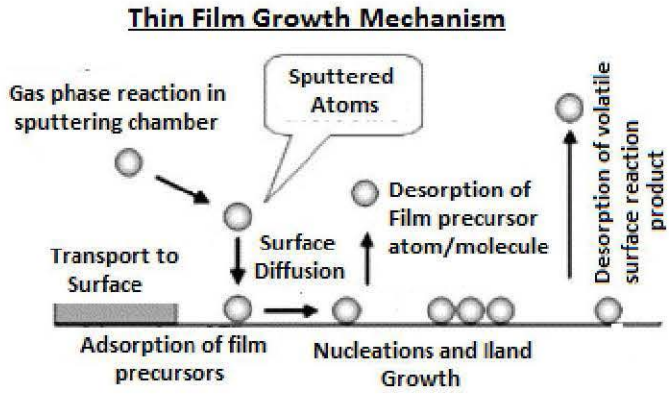


Fig. 5. Thin film formation.

depends on electrostatic forces. Polymer/ZnO adhesion is driven by the electrostatic interactions between backbones and substrates. Lattice matching of coating materials with substrate surface plays crucial role for uniform stress free film formation. On planar ZnO substrates, though some disorder is present, the polymer films can be crystalline with predominance of the face-on molecular orientation. For solar cell application doped ZnO, SnO_2:F or ITO thin film are mainly developed on glass substrate at different substrate temperature. Magnetron sputtering offers high quality crystalline films with high sticking co-efficient at higher substrate temperature (T_s). At low substrate temperature multiple nucleation sites exist at the initial stage of ZnO:Al film formation and so columnar growth are noticeably observed. At high substrate temperature adatoms gets sufficient energy for aggregation to form compact structure. Crystalline Si are also used as substrate materials for TCO coating development for different types of buffer layer, AR-coating and many other opto-electronic devices fabrications[26–30]. Lattice matching as well as substrate heating provides faster growth TCO coating that makes the process commercially and economically viable.

3. Electrical characteristics of magnetron sputtered TCO films (Electrical reistivity, sheet resistance, carrier density, mobility, work function etc.)

Electrical conduction mechanism of transparent conducting oxide thin film is based on drift of free carriers and it follows

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Drude free electron model with in the polycrystalline solid thin film. Electrical conductivity of TCO films is highly dependent on carrier density, carrier mobility, surface roughness, defects and crystallographic structure etc. Details of the contribution of each parameter in controlling the electrical properties have been discussed on the basis of experimental outcome.

Systematic studies have been done by various research groups to achieve the uniform TCO film with low sheet resistance and high visible transmittance with proper surface texture. To achieve the best electrical and optical properties of TCO films, various parameters such as, chamber pressure, RF/DC power, substrate to target distance, gas ambient, substrate temperature, dopants, etc.[31,32] are optimized. Numerous investigations have been done on the electrical properties of TCO films to understand the conduction mechanism[33,34].

In India Chopra *et al.*[34] in IIT-Delhi and Ray *et al.*[35] in IACS, Kolkata did extensive work on TCO thin film during 1990 to 2000 and reported the systematic variation of electrical resistivity, carrier concentration and mobility of magnetron sputtered TCO thin films with various deposition parameters. Thickness-dependent properties of indium-doped ZnO films are reported by Chopra *et al.*[36]. Pioneer work on ITO thin film development is done by Dutta and Ray[37] and reported $\sim 10^{-5}$ ohm-cm resistivity, sheet resistance < 5 ohm/square and carrier concentration $10^{22}/cm^3$. Later Das *et al.*[23] studied the effect of gas ambient on electrical, optical and morphological properties of magnetron sputtered AZO thin film and reported 2.8×10^{-4} ohm-cm resistivity and sheet resistance 3.5 ohm/square. Details of the variation of resistivity, carrier concentration, mobility, optical properties (bandgap, % transmittance, % reflectance etc.), structure and surface-morphology-topography with rf-power, chamber pressure, flow variations of Ar, $Ar+O_2$ and $Ar+H_2$ have been reported. Finally optimized TCO films are used as back-reflector at the Si/Al interface and significant improvement is observed in photovoltaic performance of thin film Si solar cell[38,39].

Several research groups have reported on the systematic variations of $Ar/O_2/H_2$ on the development of transparent conducting ZnO films. Minho Joo. *et al.*[40] shows the increase of crystallinity at higher oxygen pressures and the optimum resis-

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tivity is 3.7 × 10^{-4} Ω-cm and sheet resistance of the seed layer deposited under the conditions of Ar/O_2:9/1 was ~100 k Ω □. Here, resistivity decreases with increasing oxygen partial pressure.

In another report W. W. Wanga *et al.*[41] reported the variation of electrical properties of ZnO:Al thin film on the O_2/Ar flow ratio, as the ratio of O_2 increases the conductivity is improved. The lowest resistivity (**1.80 × 10^{-4} Ω-cm**) and highest carrier concentration (1.27 × $10^{21}/cm^3$) are obtained under O_2/Ar flow ratio 10:40 using Zn:Al metal target. With very low oxygen precursor zinc atoms remain un-oxidized, form colloidal zinc and aluminium oxide. The colloidal Zn is surrounded by the aluminium oxide. That is why the film prepared Zn:Al metallic target shows an extraordinary low electrical conductivity. The hexagonal ZnO is formed when oxygen content increases and the Al^{3+} ion replaces Zn^{2+} and creates more free electrons, which results the improvement of electrical conductivity.

Effect of hydrogen in controlling the materials properties of ZnO:Al films are elaborately studied by Das *et al.*[3]. Hydrogen acts as reducing agent, creates more oxygen vacancies which may cause for the increase of carrier concentration in AZO films. Under optimized condition, maximum carrier concentration and Hall mobility, were $2.3\times10^{21}/cm^3$ and 44.4 $cm^2/V.s$ respectively. The main factor of increasing carrier concentration and mobility is found to be related to hydrogen incorporation and effective substitution of Zn^{2+} sites by Al^{3+}.

Hideaki Agura *et al.*[42] developed approximately 280 nm thick-AZO films on glass substrate by PLD using an ArF excimer laser and reported the lowest resistivity of 8.54 × 10^{-5} Ω-cm with average transmittance more than 88% in the visible range. In this case, a series of AZO thin films were grown at a target-to-substrate distance of 25 mm in transverse magnetic field. Bin- Zong Dong *et al.*[43] deposited a set of polycrystalline AZO samples with different thicknesses using PLD and reported thickness dependence on structural, electrical and optical properties of AZO films elsewhere[43].

Kluth *et al.* in[44,45] IPV, Germany prepared AZO thin films by using both metallic and ceramic targets [Zn and ZnO:Al] by both DC-and Rf-Magnetron sputtering using Ar, $Ar+O_2$ and

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Ar+water vapor as sputtering ambient and did rigorous study on electrical, optical, micro-structural and surface texture. Resistivity $\sim 10^{-4}$ ohm-cm sheet resistance ~ 20 ohm/square, average % transmittance $\sim 85\%$ are reported by their group. They are trying to control the surface texture by wet-chemical as well as plasma etching for fine tuning of the surface roughness to achieve better light scattering. The effect of newly developed ZnO:Al films have been used as front electrode in thin film amorphous/miro-or nano-crystalline silicon solar cell and systematic improvement is achieved but not comparable to the commercial Asahi U-type SnO_2:F coated TCO substrate. Similar kinds of result ($\rho \sim 2.7 \times 10^{-4}$ Ω-cm at substrate temperature T_s = 250°C) for aluminum-doped ZnO (AZO) films deposited by d.c. magnetron sputtering are reported by T.Minami *et al.*[46]. Wei Liu Lu *et al.*[47]. They prepared the ZnO:Al thin film with an electrical resistivity of 5.47×10^{-4} Ω-cm, carrier concentration of 3.98×10^{20} cm^{-3} and mobility of 28.7 $cm^2 v^{-1} s^{-1}$ under Rf-magnetron sputtering on glass substrate. W.T. Yen *et al.*[48] reported the resistivity of 8.5×10^{-4} Ω-cm of ZnO:Al thin film deposited by pulsed DC magnetron sputtering.

H. Agura *et al.*[42] reported the lowest resistivity 8.54×10^{-4} Ω-cm of AZO films deposited by pulsed laser deposition (PLD) technique. Y. Igasaki and H. Kanma[50] and others[51] reported deposited AZO films using AZO (98 wt% ZnO+ 2 wt% Al_2O_3) ceramic target by rf magnetron sputtering and reported electrical resistivity 4.6×10^{-4} Ω-cm, sheet resistance of 32 ohm/square) on glass and Si wafer substrate at 250°C substrate temperature[51] and $\sim 2.5 \times 10^{-4}$ Ω-cm on amorphous substrate at substrate temperature 200°C[50] respectively. Importance of Reactive sputtering for the development of TCO films with high growth rate, high visible transmittance as well as a high conductivity for large area deposition using metallic targets is reported somewhere[52]. The growth temperature plays a major role for the determination of thin film micro-structure and opto-electronic properties.

In another report, Fang-Hsing Wang *et al.*[53] studied the effect of H_2 plasma treatment on AZO thin film and it was shown that the electrical resistivity was decreased from 1.23×10^{-3} Ω-cm to 8.23×10^{-4} Ω-cm. W. Bin. *et al.*[54] prepared ZnO:Al

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films at optimized deposition condition, with dopants concentration of about 1.2 wt% Al. Different groups are doped different element in ZnO film, such as B, Al, Ga, In, F. Among them most studied are ZnO:Al and ZnO:Ga. The electrical property of the prepared AZO and GZO films depends on the position of the dopants atoms in lattice network and solubility. In case of the TCO films prepared by spray pyrolysis technique, In and Ga is more efficiently contribute in electrical conductivity. In & Ga doped TCO shows greater electrical mobility than that of Al deped TCO films since the Al atoms are preferably lie at interstitial sites of the ZnO lattice and decrease the electrical mobility.

The systematic variation of electrical properties with doping concentration is abbreviated in the Table 2 shown below. The film shows good poly-crystalline structure with random orientations of crystallite and different grain size distribution. At high carrier concentration due to replacement of Zn by dopant atoms

Table 2. Variation of electrical properties with doping concentration

ZnO thin film sample	Deposition method	Composition of doping element (wt%)	Electrical properties (Resistivity in Ohm-cm)/Sheet resistance in Ohm/Square	% Optical transmittance	Ref.
ZnO:Ga	Sputtering	2.5	10.7	81‘.65	[100]
(GZO)		3	4.9×10^{-4}, 4.86	85	[101]
		3	3.9×10^{-4}, 4.6	85	[102]
		3.2	5.7	81.7	[100]
		4	3.4×10^{-4}	90	[103]
		4	2.61×10^{-4}	90	[104]
		5	4.28×10^{-4}, 8.5	>90	[105]
ZnO:Al	Sputtering	0.2	6.8×10^{-4}, 4.5	80	[106]
(AZO)		0.5	4×10^{-4}, 10	80	[107]
		1	5×10^{-4},	80	[108]
		1.5	7.6×10^{-4}, 10	86	[109]
		2	8.23×10^{-4}, 2.4	91.7	[110]
		2.3	3.3×10^{-4}, 5.3		[111]
		2.7	2.8×10^{-4}, 3.2		[112]
		3	4.7×10^{-4}, 3.1	90	[108]
		5	10^{-3}	80	[108]

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from ZnO lattice site, the ionized impurity scattering predominates over grain boundary scattering. The carrier mobility directly depends on the carrier density by the formula $\mu\alpha N^{-2/3}$, at low doping levels. Reactive sputtering using metallic targets is the most promising technique because of its high deposition rate, large area scalability, and easy preparation of a large size as well as a high conductivity and visible transmittance[52].

In 2013, Bo Ho *et al.* also shows the variation of substrate temperature makes the film were oriented more preferentially along the (002) direction, the grain size of the films increased and the electrical resistivity decreased. The resistivity of 4.973 $\times 10^{-4}$ ohm-cm at substrate temperature 480°C, with the high carrier concentration and mobility as 1.103 $\times 10^{21}$ atom/cm^3 & 11.4 cm^2/VS respectively[53].

The growth temperature plays a major role for the determination of thin film properties. It is observed that upto T_s = 300°C the resistivity of ZnO:Al film is of the order of $\mathbf{10^{-4}}$ **Ω**-cm. At higher temperature oxygen desorption is much higher, which results in decreasing the electron traps. Initially, with the increase of substrate temperature results in bigger grains size and less numbers of defects, which ultimately makes improvement of crystallinity. Hence, resistivity decreases due to increase of both carrier concentrations and Hall motilities[54] **at** intermediate substrate temperature, while for further increase of substrate temperature, the resistivity of the films is increased due to decrease of oxygen defect resulting in the decrease in carrier concentration.

Sung Ju Tark *et al.*[55] also reported that the variation of resistivity of AZO film in rf magnetron sputtering system with substrate temperature. The resistivity of the AZO:H films improved from 4.98 $\times 10^{-3}$ to 3.21 $\times 10^{-4}$ Ω-cm with of as the substrate temperature varies from RT to 150°C. This result was apperently due to the increase in the substrate temperature, which accelerated the crystallization of the AZO:H films and the incorporation of hydrogen atoms into O vacancies, thereby decreasing the concentration of interstisital impurities in the crystal and slightly decreasing the resistivity and increasing the mobility, respectively. Also they notice that the hydrogen makes O-H bonds with oxygen through the doping, incfreases the number of O vacancies and function as a shallow donor by

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filling O vacancies. As the substrate temperature was increased, the energy increased as well, which improved the resistivity. This result can also be explained by the decrease in the lattice distortion. The decrease in the resistivity with increasing substrate temperature can be explained firstly by the increased electric charge carrier due to the addittion of Al_2O_3 (2 wt %), which caused the replacement of Zn^{2+} ions by Al^{3+} ion and secondly by the combination of the doped hydrogen atoms with oxygen atoms, thereby creating OH^- bases and the increase in O vacancies which were replaced by hydrogen atoms to be shallow doners. They give a conclution their was no significant change in the resistivity and the carrier concentration with increasing substrate temperature from RT to 250°C after deposition of the thin film. However, the mobility increased in greatly from 12.39 to 21.32 cm^2/vs, which improved the electrical properties.

The effect of substrate temperature, on electrical resistivity of AZO thin films were reported by Jinsu Yoo *et al.*[56]. The electrical resistivity of the etched ZnO: Al films have been compared with the smooth film, and the values are shown in Table 2. AZO films deposited at an argon pressure of 0.27 Pa and at 573 K exhibit very low electrical resistivity (1.9×10^{-4} Ω-cm). Kang Hyon Ri *et al.*[57] achieved the low resistivity of 7.1×10^{-4} Ω-cm with the variation of substrate temperature in rf sputtering technique.

Very recently Piu Chuan *et al.*[58] reported the effect of variation of substrate temperature of AZOY thin film. The AZOY film reach the minimum resistivity, deposition on soda-lime glass substrates by DC magnetron sputtering of 4.26×10^{-4} Ω-cm at T_s = 400°C, with maximum value of carrier concentration of 1.86×10^{21} cm^{-3}. The substrate heating is detrimental to the surface morphology which can affect the film resistivity. Higher surface roughness leads to a non uniform morphology of the film[41]. The films with more defects may face stronger grain boundary scattering which ultimately results in shrinkage of carrier lifetime. The combined effect of high surface roughness, more void and defects in thin film structure may be the cause of low mobility and high resistivity of thin film TCO materials. The minimum resistivity achieved for film deposited at T_s = 400°C which has compact surface structure, good crys-

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tallinity, larger crystal grains and small RMS roughness, simultaneously.

Electrical resistivity, sheet resistance, film uniformity and sticking co-efficient are strongly dependent on chamber geometry, rf-power, chamber pressure, substrate temperature and substrate –target distance. At low substrate temperature TCO thin film becomes more resistive but at high substrate temperature sample resistivity is optimum whereas at excessive high substrate temperature the film becomes more resistive. With increase of rf-power and at low pressure sputtered TCO thin films generally becomes more conducting and otherwise film property start to deteriorate.

M. K. Jayrajm *et al.*[59] reported the resistivity of ZnO:Al films (ρ - 4.5 × 10^{-4} Ω-cm) deposited under off axis method at optimized condition at 423 k substrate temperature. At high temperature, film exhibited higher doping efficiency, higher carrier concentration, lower resistivity, and higher hall mobility. Basically low growth temperature is required for the flexible substrate in OLED. Only drawback in off-axis configuration results in non-uniform film thickness across the substrate.

Based on application point of view, thickness dependence of electrical properties of TCO films is another big issue. Systematic variations of resistivity with the variation of thickness Al doped ZnO thin films deposited by different deposition techniques have been reported (shown in Table 3). ZnO –based TCO films shows versatile applications depending upon film thickness which changes from ~50 nm to ~5000 nm. In CVD, PLD and magnetron sputtering technique, ZnO films thickness can be controlled with the variation of deposition time, substrate temperature, chamber pressure, rf-power and chamber geometry[60,61]. To improve the resistivity of films, substrate to target distance plays a significant role. M. K. Jayrajm *et al.*[59] reported the variation of deposition rate with substrate to target distance. Thickness dependent TCO film properties prepared by sol-gel, spin coating techniques were reported somewhere[62–64] and TCO films thickness were varied with different repeating times of coating procedure.

Kang hyon Ri *et al.*[57] shows the thickness dependence of resistivity in AZO films. The resistivity of TCO films decreases rapidly with increasing of thickness in both case of 100°C and

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350°C. The lowest resistivity was about 3.14 × 10^{-3} ohm-cm for those films with thickness range of 500–600 nm deposited at substrate temperature 100°C and 7.1 × 10^{-4} ohm-cm for films thickness range 350–400 nm when deposited at 350°C.

Table 3. Thickness dependent electrical and optical properties

Sample	Film thickness (nm)	Resistivity (Ohm-cm)	Sheet resistance (Ohm/Square)	% Optical transmission	Ref.
RF-Sputtered	100	6.11×10^{-4}	61.1	92.8	[65]
ZnO thin film	143	4.6×10^{-4}	32.16		[66]
	260	3.21×10^{-4}	12.3	86	[55]
	300	5.0×10^{-4}	16.67	85	[67]
	330	8.23×10^{-4}	24.9	91.7	[68]
	395	5.14×10^{-4}	13	86	[69]
	440	5.1×10^{-4}	11.59	80	[73]
	530	4.0×10^{-4}	7.5	89	[54]
	592	7.1×10^{-4}	11.9		[70]
	800	2.8×10^{-4}	3.5	90	[3]
	837	1.8×10^{-4}	2.1	85	[41]
	1195	1.9×10^{-4}	1.59	>85	[56]
	1500	6.8×10^{-4}	4.5	80	[71]
	2000	1.32×10^{-4}	0.66	87.9	[72]

Recently Hung-wei. Wu *et al.*[65] reported the AZO thin films prepared in Rf sputtering techniques with different Ar flow (from 30–60 sccm). The lowest resistivity of 6.11 × 10^{-4} ohm-cm achieved under optimum RF power of 250 W, Ar flow of 40 sccm. The grain size of AZO thin films increases with the increase of Ar flow upto 40 sccm. Large grain structure enhances crystallinity and carrier lifetime[67,68] and may reduce the lattice defects and grain boundary scattering, which may cause enhanced mobility. The increase in both carrier concentration and Hall mobility would consequently lead to an increase of conductivity for AZO thin film deposited at Ar flow of 40 sccm under sputtering power 250 W. The resistivity of AZO thin films increases when it is grown at higher Ar flow above 40 sccm. Further increase of Ar pressure in sputtering chamber may cause the incorporation of Ar^+ into the AZO thin films and glass substrate becomes contaminated. The above facts may

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be the possible reason for the enhanced resistivity at higher deposition pressure[66].

The electrical resistivity of ZnO:Al films is lower than that of the undoped ZnO films and it is due to the free electrons released by the substitution of Zn^{+2} ions with Al^{+3} ions at the lattice sites. **Several groups** have studied the variations of electrical and optical properties of ZnO:Al films deposited on different substrates, such as glass, sapphire, and polymeric substrates [polyethylene-terephalate (PET)[69], polycarbonate (PC)[70], as polyphalamide (PPA) *etc*]. Due to the unique flexible nature, light weight, and low cost of polymer substrates, TCO thin film coated plastic substrate has promising application in flat panel display like OLED and other kinds of flexible electronics[85,86].

Very recently focus has been given by the researcher for further development of Ga/In/Al doped ZnO based alternative TCO thin film development for commercial applications. Among the various preparation methods, pulsed laser deposition is more appropriate technique for precise controlling electrical, optical, and morphological properties as well as the stoichiometry.

In 2005, Xuhu Yu *et al.*[102] developed n-type ZnO:Ga (GZO) thin film by rf-sputtering and studied the effect of rf-power variation on electrical properties of GZO film. They reported lowest resistivity of 3.9×10^{-4} Ohm-cm, sheet resistance 4.6 ohm/square for 800 nm thick GZO film. In 2006, Park *et al.*[1] studied the effect of substrate temperature (from 100°C to 400°C) on the properties of GZO film prepared at an oxygen pressure of 0.67 Pa and reported lowest electrical resistivity of **8.12×10^{-5} Ω-cm**, a carrier concentration of 1.46×10^{22} cm^{-3}, carrier mobility of 30.96 cm^3/Vs and % visible transmittance (T) ~90% in Pulsed Laser deposition (PLD) technique.

In 2010, Tae Youn Kim *et al.*[105] studied the effect of substrate temperature variation (T_s varies from room temperature to 300°C) and doping concentration on the properties of the **GZO films by DC magnetron puttering technique.** A minimum value of **4.28×10^{-4} ohm-cm** and a high transmittance (> 90%) is obtained at 300°C with film thickness 500 nm. The Ga concentration of all GZO films is about 4 wt%, but the film prepared at 200°C has a higher Ga concentration than the film

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CO films.

prepared at 300°C. The higher Ga concentration in ZnO:Ga thin film results in an increase of resistivity due to carrier recombination in grain boundary regions.

In 2012, *Han-Ki Kim et al.*[38] reported the thickness dependent electrical properties of cylindrical rotating magnetron sputtering **(CRMS)**-grown GZO films. The sheet resistance (R_{sh}) of GZO film were respectively 137.7 Ohm/square and 4.86 Ohm/square for film thickness 50 nm and 1,000 nm. The carrier concentration and mobility of the GZO films increase with increasing thickness from 50 nm to 1,000 nm. In last 10 years, Agura *et al.*[42], Park *et al.*[1] and Christopher W. *et al.*[74] reported the latest low resistive AZO and GZO thin films samples deposited on glass substrate with very low resistivity 8.54×10^{-5} Ω-cm(AZO), 8.12×10^{-5} Ω-cm(GZO), and 7.7×10^{-5} Ω-cm (GZO) respectively. Very recently Das *et al.* (3) developed ultrathin IZO:Ga and ZnO:Al on a soda lime glass and plastic substrate at room temperature with resistivity 10^{-4} and 10^{-5} ohm-cm, 82% & 90% optical transmittance. P. Gondoni *et al.*[75] reported lowest electrical resistivity of 4.89×10^{-4} ohm-cm with a Hall mobility of 25.9 $cm^2 V^{-1} s^{-1}$ of sputtered ZnO:Ga (97:3 wt%) thin film.

Mainly In^{3+}, Al^{3+}, B^{3+}, Ga^{3+} are commonly used n-type dopants in ZnO[76–80]. Most of the researchers prefer to work with Al doped ZnO. As the ionic radius of Al^{3+} is smaller than Zn- ion (ionic radii of Al^{3+} and Zn^{2+} are 0.54 Å & 0.74 Å respectively), Al^{3+} can easily occupy the place of Zn^{2+} in lattice, leading to a reduction of the lattice parameter. Also the covalent bond-lengths of Ga-O and ZnO are 1.92 Å and 1.97 Å, respectively reported somewhere[81]. Both Ga^{3+} and Zn^{2+} ions has similar ionic radius and for that reason Ga doped ZnO has minimum lattice deformations even at higher Ga-doping concentrations[82,83]. On the other hand, Ga is less reactive and more resistive to oxidation compared to Al[84]. This may explain the better merits of Ga-doped Zinc oxide films (ZnO:Ga) compared with Al-doped Zinc oxide films (ZnO:Al) in terms of electrical properties of doped ZnO-based TCO films[22]. Carrier mobility in polycrystalline films can be significantly influenced by gain boundary scattering, scattering centres as well as on surface roughness scattering. The crystallographic defects (grain boundaries, dislocations, interstitials and vacancies) induce the scattering of carriers and carriers can be trapped.

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4. Optical properties

The optical property of TCO films is mainly characterized by its transmittance, reflectance and absorbance. In general TCOs have a transmission window between the wavelengths of about 400 nm and 1500 nm. The short wavelength (UV) cut-off corresponds to the fundamental bandgap of the material, whereas Drude free electron model explains the free carrier plasma resonance in the longer wavelength (IR-edge) side. Experimentally optical transmittance (T) and reflectance (R) can be measured, but the absorbance (A) can be experimentally determined with the help of following relation $A(\lambda) = 1 - T(\lambda) - R(\lambda)$. The frequency dependent dielectric function (ε) can be expressed in terms of permittivity and carrier density mathematically given by

$$\varepsilon(\omega) = 1 - \frac{ne^2}{\varepsilon_\infty \varepsilon_0 m_e^* \omega^2} \Rightarrow \varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2} \quad (1)$$

Where ε_∞ is the high-frequency dielectric constant (for a ZnO thin-film

$\varepsilon_\infty = 3.61$). The plasma frequency (ω_p) is defined as

$$\omega_p = \sqrt{\frac{ne^2}{\varepsilon_\infty \varepsilon_0 m_e^*}} \quad (2)$$

These critical optical properties are directly influenced by the carrier density and mobility. The intrinsic properties of transparent conducting thin films like optical transmittance (T), reflectance (R), and absorbance (A) are determined by its refractive index n, extinction coefficient k, bandgap E_g but on the other hand sputtering chamber design and geometry (extrinsic) plays crucial role for TCO film thickness control, film uniformity, adhesion and film surface roughness. Energy bandgap (E_g) depends on the chemical composition and solid structure of the material. For TCO films, E_g increases at larger carrier density, known as Burstein-Moss effect. There is a negative correlation between the carrier density and the position of the IR absorption edge, but positive correlation between the carrier density and the UV absorption edge. As a

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result, the TCO transmission boundaries and conductivity are interconnected.

Fig. 6 shows the variations of optical transmittances as well as reflectance of AZO thin films as a function of flow rate of different gas ambient. $A+H_2$ deposited rf-sputtered AZO films provide an excellent UV cut-off and IR shielding effect with 95% average visible transmittance. This characteristic can be understood from the fact that the AZO is a direct bandgap semiconductor with an energy gap of ~3.2 eV[85].

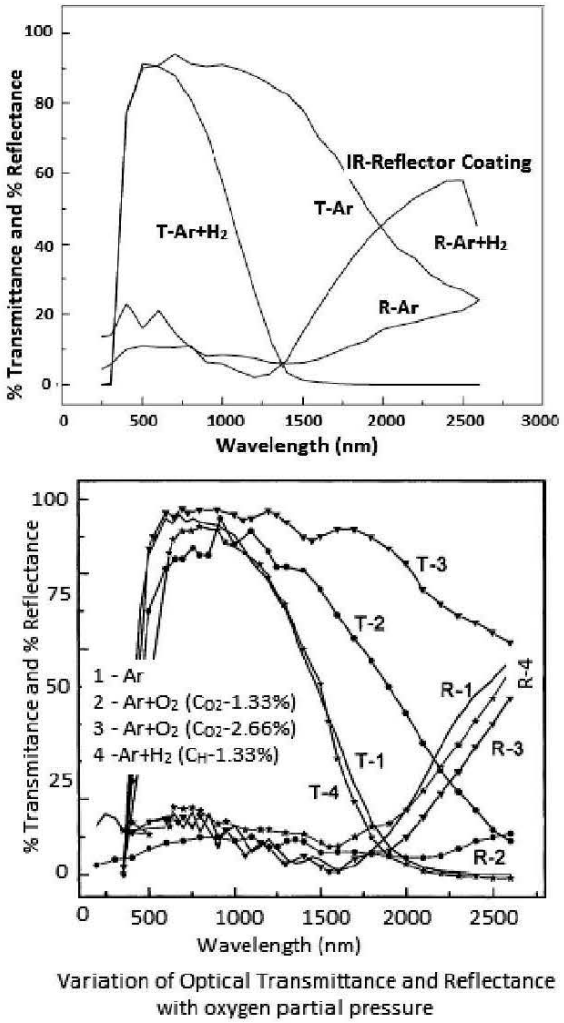


Fig. 6. Optical transmittance and reflectance spectra of TCO films.

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The width of the VIS transmission window of a TCO film deposited on a transparent substrate is affected not only by the optical parameters of the TCO film but also by the optical properties of the substrate. The refractive indices of the most common substrates (n_{sub}) are ~1.45 and 1.6 respectively for fused silica and various glasses. The extinction coefficient of the substrate (K_{sub}) is generally $< 10^{-7}$, hence any light absorption would take place in the film, where generally

$K_{film} > K_{sub}$.

Several interference bands are observed for thicker films (film thickness greater than 100 nm) with the variations of wavelength or thickness producing maximal and minimal values of T. Hence, assuming that the sample is in air, T_{max} = 90 and 93% for films deposited on glass and fused silica respectively, the minimum sample transmission (T_{min}) in air is expressed by:

$$T_{min} = \frac{4n^2 n_{sub}}{(1+n^2)(n^2+n_{sub}^2)} \quad (3)$$

The refractive index (n) of most TCO films is closely related by 1.8 - 2.8 T_{min} in the VIS in the range, where T_{min} is approximated by the relation:

$T_{min} = 0.051\ n^2 - 0.545\ n + 1.654$.

With decrease of TCO film thickness, T increases but the sheet resistance (R_s) decreases. Combining together the optical and electrical properties of the thin solid film, the fraction of the flux absorbed in a film (A) is given by the expression:

$$A \cong \left(1 - \exp\left(\frac{-\alpha}{\sigma R}\right)\right) \quad (4)$$

X. Bie *et al.*[103] observed the optical transmittance of the ZnO:Ga films at various temperatures in the visible wavelength range.

It is evident that there is an optimum O_2 concentration for the deposition of IZO films with low resistivity and high transmittance. It can be explained as the deposition of stoichiometric IZO films due to the compensation for oxygen vacancy. The observation of the films by AFM indicated that

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The microstructure and surface morphology of IZO films play key role in controlling the optical transmittance. IZO thin film is one of the new promising TCOs, and is likely to be suitable for low temperature process using flexible substrates, such as flexible displays and solar cell[63].

The band gap of the ZnO: Ga films undergo a blue shift due to increase in carrier concentration that leads to a filling of the lower energy states of ZnO near the conduction band minimum. Hence, there is a shift in the absorption edge to higher energies, known as the Burstein-Moss (BM) effect. It is observed that AZO:H films deposited using AZO (2 wt%) target under various substrate temperature and in reactive plasma environment (hydrogen and Ar precursor gas) show optimum the visible transmittance 85 - 86%, in the wavelength range of 250 - 1100 nm and it shows better optical characteristic for C-axis orientation Hydrogen incorporation should increase the diffusion of light due to surface texturing.

The visible transmittance vs wavelength for the samples (sample thickness ranged from 250 to 800 nm) deposited at different sputtering power for sputtering pressure 1 Pa and sputtering time 20 min reported by Xuhu Yua. *et al.*[102]. It is seen that the average transmittance of these samples in the range 400–800 nm was over 85%. The optical gap (E_g = 3.6 eV) of the films were obtained by plotting (α^2 vs $h\nu$) and extrapolating the straight line portion of this plot to the energy axis shown in the Fig. 6.

Han-Ki Kim. *et al.*[56] prepared gallium doped zinc oxide (GZO) films by cylindrical rotating magnetron sputtering (CRMS) and reported the variation of optical transmittance with film thickness under optimized deposition conditions. Interference pattern was observed with increase of GZO film thickness, and 250 nm thick GZO film shows highest visible optical transmittance (% T) of 89.97% (wavelength range between 400–800 nm) but % T drastically decreases in the near infra-red range (above 900 nm) due to free carrier absorption. With increasing film thickness the decrease of optical transmittance of GZO thin film may be explained according to Drude model due increased electron-photon interaction. Film thickness plays crucial role to control the electrical and optical properties in ZnO-based TCO films. DC magnetron sputtered zinc oxide

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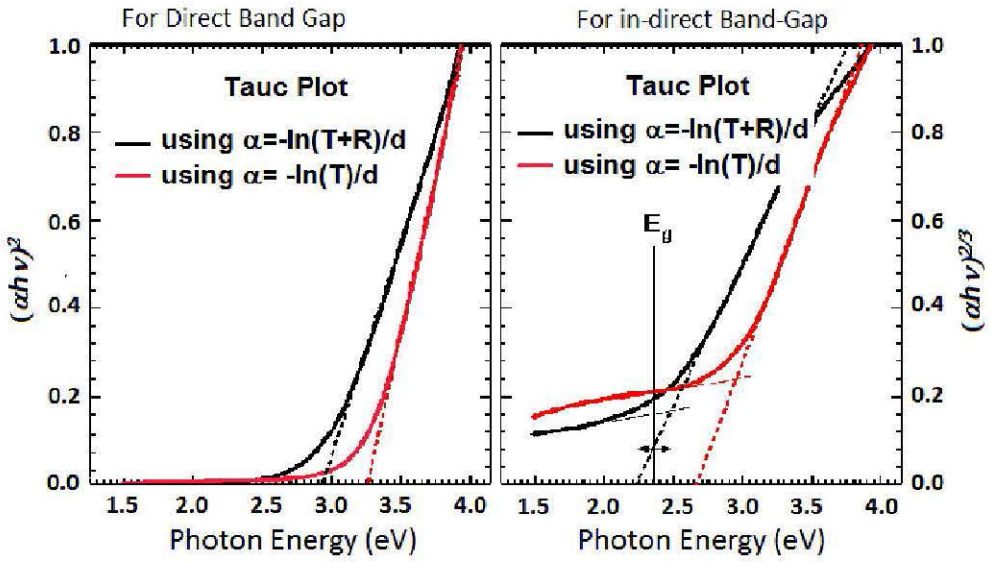


Fig. 7. Tauc Plot for determination of Bandgap of TCO coating.

(ZnO) films with (002)-orientation columnar structure grown in pure Ar but granular structure in H_2O-Ar mixtures at elevated substrate temperatures on glass substrates.

5. Effect of surface texture and haze factor

Proper Surface texture of TCO thin film plays crucial role for effective light trapping in Tandem solar cell. Optimum surface texture/morphology can be achieved by proper tuning of etching parameters during wet chemical etching as well as reactive plasma etching. Surface roughness and Haze factor are controlled by fine tuning of etching time, and other relevant etching parameters. Under optimum condition for both wet chemical as well as dry etching, with increase of etching time surface roughness increases and as a result specular transmission decreases but diffuse scattering predominates.

The total reflection of light from a surface of normally distributed height is given by

$$R_\theta = R_0 \exp\left[\frac{-4\pi\delta}{\lambda}\right]^2 + \left(\frac{32R_0}{m^2}\right)\left(\frac{\pi\delta}{\lambda}\right)^4 (\Delta\theta)^2$$

where, λ is the wavelength of light, δ is RMS Roughness, m is

mean gradient of the surface, R_0 is the Reflectivity of the coating and $\Delta\theta$ is the solid angle.

In the above expression the 1st and 2nd term corresponds to specular and diffuse scattering respectively. When the surface roughness (δ) increases, diffuse scattering predominates. Ar deposited sputtered ZnO:Al films have mostly flat surface, and both the specular as well as partial diffuse scattering mechanism co-exist. As deposited ZnO:Al films under various hydrogen dilution ratio shows different haze factor and ultimately affects light scattering mechanism.

Proper surface textures of TCO thin film can play significant role in enhanced light scattering as well as the effective light absorption in intrinsic layer of the solar cell due to multiple reflections inside the active layer of thin film based solar cell. Apart from that TCO surface texture can minimize surface recombination losses, ultimately, improves the efficiency and stability of thin film solar cell.

Two novel methods namely dry etching (by plasma treatment or Reactive Ion Etching (RIE)) and wet-chemical etching (using different acids with suitable concentration for different durations) were applied for fine tuning to achieve proper surface texture of TCO thin films. The haze factor of TCO films (for reflection and transmission) defined by diffuse to total light scattering from TCO surface (expressed as shown in the following equations $H_R = \frac{R_{diff}}{R_{total}}$ and $H_T = \frac{T_{diff}}{T_{total}}$ were significantly controlled by varying the etchant species, acid concentration, and etching time.

According to scalar scattering theory, the commonly used analytic function for the haze in reflection related to surface roughness (δ) and the wavelength (λ) of incident light can be expressed as

$$H_R = 1 - \exp\left[-\left(\frac{4\pi\delta}{\lambda}\right)^2\right].$$

The above formula is valid under following assumptions (I) the scattering surface is perfectly conducting and (II) $\delta/\lambda << 1$.

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The above scattering formula will be applicable for ZnO surfaces with a thin (200 nm) silver layer coating on glass/ZnO substrates surface for the wavelengths 800 nm $< \lambda <$ 1100 nm.

Control of surface texture

Sputtering gas plays crucial role for fine tuning of surface morphology, micro-structure, internal texture, grain size distribution as well as optical properties. Especially the effect of hydrogen on ZnO:Al films properties are very much interesting. The variation of hydrogen concentration C_H ($C_H = H_2/(Ar+H_2)\times 100\%$) in plasma environment controls the grain distribution, grain size variation, surface texture and haze factor of as deposited ZnO thin film. As the hydrogen dilution increases deposition rate decreases and grain size also decreases. Haze factor is highly correlated with variation of grain size of TCO film. It is observed that diffused transmittances initially increases then decreases with increase in hydrogen concentration in vacuum chamber during film deposition and finally saturates. Rf-sputtered ZnO films under reactive ($Ar+H_2$ ambient) environment with C_H = 5% and 10%, the surface morphology becomes more rough and porous. As a result, Haze factor of non-stoichiometric ZnO:Al films increases from 36% to maximum 42%. Details of scattering mechanism from rough TCO surface, has been discussed elsewhere[22] and experimental investigation on Haze factor of doped textured ZnO film is corroborated with theoretical prediction. Excessive increase of H-species in reactive environment may cause for less compact non-stoichiometric ZnO thin film due to increase of both amorphous structure and void in ZnO:Al matrix. The changes in internal texture as well as micro-structure of ZnO thin films in terms of geometrical shape, distribution of size, may lead to increased absorption in longer wavelength region.

Many researchers studied the optical properties and morphologies of plasma treated Al-doped textured ZnO (AZO) and Ga–doped (GZO) films. Significant variations of surface roughness as well as haze factor are observed. H_2 plasma treatment plays significant role to enhance the opto-electronic properties of AZO thin films and figure of merit (FOM) of TCO thin film. Higher value of figure of merit (FOM) of thin film means better balance in optical and electrical properties.

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Several groups have put their interest in plasma-etching or Reactive Ion Etching (RIE) techniques to control surface texture ZnO:Al thin film. Reactive Ion Etching (RIE) has high-throughput in contrast to laser or mechanical texturing. Dry texturing technique using hydrogen radicals is mostly promising approach for fine tuning to achieve excellent surface prosperities, where the surface roughness is a basic challenge. In order to create a suitable surface texture for high efficiency solar cells, the samples surfaces needed further smoothing treatment after dry texturing. Systematic analysis of plasma etching to control Haze factor of ZnO:Al thin films has been reported elsewhere[19-21]. Das *et al.*[23,35] studied Hydrogen plasma etching on ZnO:Al films and those results are compared with commercial TCO films. In that article, comparative optical properties (total and diffuse transmission spectra) of lab based sputtered ZnO films with commercial TCO has been reported. Commercial TCO films are U-type SnO_2:F coated thin films and those films show high optical transmission (90%) over wide range of wavelength in solar spectrum. Haze factor of U-type commercial SnO_2:F films varies from 7% to maximum 25%. Laboratory based RF-sputtered ZnO:Al films under hydrogen plasma environment show almost 90% visible transmission as well as 60% NIR transmission at 1100 nm and significant amount UV-absorption loss.

Fang-Hsing Wang *et al.*[87] studied the effects of H_2 plasma treatment on properties of RF magnetron sputtered AZO thin film deposited under different substrate temperatures ranging from RT to 300°C. In this study, only slight increase of surface roughness and surface grain size alongwith improved film resistivity have been reported. H_2 plasma treatment does not show informations regarding surface texturizations as well as Haze factor. In a different experiment Fang-Hsing Wang *et al.*[87] reported surface texturization and control of haze of as deposited Al-doped ZnO thin films with working pressure 5×10^{-3} Torr at 200°C by both RIE as well as wet chemical etching. The as-deposited AZO films were consecutively H_2 plasma treated at 200°C for 30 min in a typical PECVD chamber and HF/HNO_3 mixture treated for further smoothing process. The novel mechanism for fine control of ZnO:Al coated substrates is implemented by using 5% HF solution and

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rinsed in pure water. The cleaned samples were exposed under RF hydrogen plasma and textured by generated hydrogen radicals.

Young-Hee Joo *et al.*[88] studied the etching mechanism of ZnO thin films using gas mixture of $N_2/Cl_2/Ar$ (15:16:4 sccm) gas in inductivity couple plasma and reported enhanced etching rate of ZnO thin films in $Ar+N_2$ gas ambients. At a high bias power, N ions increased the effective ion flux in plasma chamber and accelerated ion bombardment as well as physical etching mechanism with Ar ions.

Wet Chemical Etching is another sophisticated technique to control the surface texture. Diluted HCl with concentrations of 0.1%, 0.2%, and 0.5% in H_2O are conventionally used for Surface texturization of AZO films. Kluth *et al.*[89] reported detail study on wet chemical etching of ZnO thin film by diluted HCl acid for different etching time[44–46]. As-deposited ZnO films exhibits a relatively smooth granular structure compared to wet-chemically etched ZnO:Al films. HCl etched AZO films becomes more rough, and crater-like surface structure. Initially with increase of HCl concentration, the feature size increases, but excessive increase of HCl concentration, the feature of the etched surface changes from steep to flat.

J. Hüpkes *et al.*[90,91] studied wet chemical etching of room temperature (substrate temperature 50°C) deposited smooth ZnO:Al films on PEM substrate using both diluted hydrochloric acid (0.5% HCl) and potassium hydroxide (33% KOH) for 30–40 seconds. It was observed that the etch rate of the films is high (> 10 nm/sec) for low PEM intensities close to the oxide mode and decreases approaching the metallic mode to an almost constant level.

N. Senoussaoui *et al.*[92] used diluted HNO_3, HCl, and H_3PO_4 with a pH value of 1.0 and haze factor was 43%, 36%, and 26%, respectively for wet chemically treated ZnO:Al film samples (etching time 20s). Among them HNO_3 etchant gives the highest haze ratio of 43.0% at a wavelength of 550 nm. Similar kinds of treatments are reported by Tae Youn Kim *et al.*[105] and E.Bunte *et al.*[93]. E. Bunte *et al.* used diluted HF & HCl solution and reported high etching rate of ZnO:Al films deposited at different substrate temperature. The etching rate is

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smaller for high temperature deposited ZnO:Al films due to more compact structure of ZnO:Al films prepared at high substrate temperatures[94]. As HCl is a strong acid while HF is a weak acid, HCl would be completely dissociated into H^+ and Cl^- [95] whereas only a very small fraction of HF molecules are dissociated though both have same concentration (0.5%). Though for higher concentrated HF than HCl, but the etching rate is slower than HCl due to lower pH value of HF. It is reported that the topographies of films prepared at high temperatures (350°C & 325°C) show a wide but shallow craters with diameter of about 1-2 μm sporadically distributed on the surfaces. Both surface topographies and the height distribution function shows a more or less Gaussian shape indicating a statistical rough surface similar to that of kluth *et al.* reported earlier[94]. Moreover, Kluth *et al.*[96] showed higher etching rate for low substrate temperature deposited AZO films due to their low compactness.

Q. QIAO *et al.*[55] reported the optimization of process conditions for surface texturization of GZO films. For smooth GZO films, diluted 0.5% HCL solution is used as etchant for very low duration (etching time is typically around 15 s) at relatively low temperature[97] for industrial processes. For further improvement of surface texture, Q. QIAO *et al.*[100] used acetic acid and sodium acetate buffer solutions at low temperature and the flat GZO films are anisotropically etched leading to a crater-like texture surface, with rms (δ) values are 46.919 nm (60s) and 72.929 nm (90s) as determined by AFM. Y. C. Lin *et al.*[98] reported the etching techniques for pulsed direct current magnetron sputtered ZnO:Ga transparent conducting thin films and used different aqueous solutions, (0.5% HCl, 5% Oxalic acid, and 33% KOH solution) to tune the haze factor for improvement of light scattering properties of GZO films. TCO films textured in 5% Oxalic acid solutions for 75s show lowest resistivity of 4.3×10^{-4} ohm-cm with relatively high total optical transparency of T_{total} = 75.1%, as well as high Haze value of H_T = 0.3 at 300 K. The above result would be the most suitable for photovoltaic application. AZO thin film etched with 5% oxalic acid for 150 s deposited at substrate temperature T = 473 K, pressure-0.4 pa, 150 w power shows resistivity of 7.9×10^{-4} ohm-cm with high haze value H_T = 0.2[99].

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6. Role of magnetron sputtered ZnO:Al and ZnO:Ga for solar cell application

Textured TCO coated substrate plays significant role as front electrode as well as window layer in thin film solar cell. The ideal TCO satisfy the key features like high optically transparent (almost 95% transparency over a wide range of solar spectrum) with high electrical conductivity. Textured TCO surface controls light scattering, and provides improved light harvesting in the active layer in the solar cells.

The photo generation mechanism of free carriers is explained via an Optical Model (contribution from both specular and diffuse reflectance and transmittance at the interface in multilayered structure of thin film solar cell has taken into account). The light scattering mechanism and light trapping has been explained with the help of the Fig. 8 (shown previous page).

Introduction of 80 nm thick textured ZnO:Al thin film at Si-metal interface helps to increase the light scattering at a-Si/ ZnO interface, multiple reflection takes place, as a result

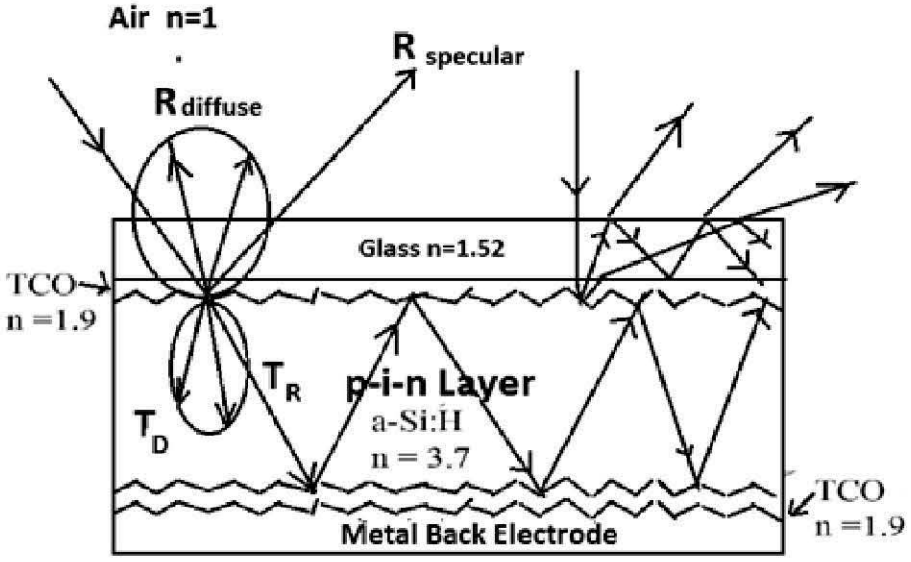


Fig. 8. Light scattering at n-layer/TCO interface and Light Trapping in Active Layer of p-i-n structure through Multiple Reflection for photo-carrier generation.

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longer wavelength response is increased and effective light trapping in the active layer in amorphous silicon solar cell via total internal reflection of weekly absorbed light. For that, moderate surface roughness [surface roughness (δ) ~40 nm for SnO_2 thin film and 50 nm for ZnO:Al thin film] is more effective to improve the short circuit current of the bottom cell due to an enhanced longer wavelength response in multijunction solar cell. Introduction of thin ZnO:Al film at rear interface can additionally can increase the electric field across the intrinsic layer in bottom shell (decreasing i - layer thickness in p-i-n structure) and ultimately improves the stability and power conversion efficiency in tandem solar cell.

Apart from doped ZnO thin film, U-type SnO_2:F is mostly used commercially for thin film amorphous silicon solar cell fabrication. In comparison to ZnO thin film, due to weak Sn-O, and In-O bond strength, ITO and SnO_2:F thin film degrades under hydrogen plasma. Micro crystalline, nano-crystalline, micro-morph thin film solar cells are generally fabricated under high hydrogen plasma. That is why doped ZnO thin film plays better performance for micro and nano crystalline even in amorphous silicon thin film solar cell.

Conclusion

Magnetron sputtering is one of the technologically important vacuum deposition techniques for thin film coating development. Magnetron sputtering plays important role for large area uniform damage free ZnO based TCO coating fulfilling low electrical sheet resistance, high optical transmittance with high figure of merit and high sticking coefficient. One can control TCO texture as well as Haze by Magnetron sputtering with the variation of sputtering conditions and gas ambient used during reactive sputtering. The Haze factor is an indicator for light scattering properties of TCO thin film. Apart from the electrical and optical properties, work function can also be tuned by changing sputtering parameters and doping profile for suitable technological applications such as organic solar cell development. Finally transparent conductive coating satisfy technical feasiability and cost effectiveness for commercially viable technological development.

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Prof. Das has established Materials Science laboratory (Research Grant: Rs. **96.72 lakhs, Funding Agency: DST and CSIR, Govt. of India**) in Haldia Institute of Technology, West Bengal. His field of research is to development of transparent conducting thin films, optical coating, perovskite materials, quantum dot phosphor etc. One student is already Ph.D awarded under his supervision.

Prof. Das has organized national conference/seminar/staff development programme and delivered invited lecturers as resource person in different organizations. He has published more than 60 research articles, book chapters in reputed journals and conference proceedings.

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